NEISS-AI · HAZARD WATCH

Which products started hurting more people in 2025?

Subash Kumar
Independent road safety and injury researcher
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NEISS-AI now checks **every product code** in the U.S. national emergency room injury data (about 800 products, 2019–2025), not only products someone already suspects. It flags a product when its injuries rose clearly faster than all emergency visits, after strict checks for chance, small numbers and coding changes. It also reads the injury stories for **new words** that start appearing more often.

2025 signals (compared with 2024)

PRODUCT	2024	2025	CHANGE	PER 100,000	STATUS	WHAT IT MEANS
E-bikes	59,600	102,600	+72%	17.5 → 30.0	SIGNAL	Largest real rise. Admissions also rose (+75%), but too few cases to pass the strict test alone.
Powered scooters (e-scooters)	118,500	165,700	+40%	34.9 → 48.5	SIGNAL	Second year of fast growth in injury-story wording (“e-scooter”).
Power workshop grinders, buffers, polishers	10,200	18,800	+85%	3.0 → 5.5	SIGNAL	New signal, spread across many hospitals. Worth a closer look at the stories.
Infant and toddler play centers	400	1,600	+304%	0.12 → 0.48	CHECK	Small numbers (49 sampled cases in 2025). Could be a real rise or a coding change.
Floors or flooring	1.86 M	3.08 M	+66%	546 → 901	CODING	Jump starts suddenly in April 2025 at most hospitals: most likely a change in how falls are coded, not a real rise.

National estimates of emergency department visits (CPSC NEISS sample weights). Per 100,000 = visits per 100,000 US residents (Census Bureau population estimates). Each flag passed a false-discovery check across all products. Full results, with 95% uncertainty and every product, are in the downloadable tables.

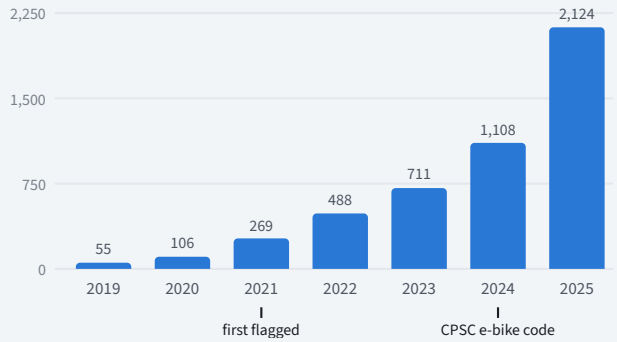
Would it have caught e-bikes early?

Yes, through the injury stories. Words such as “electric bike” and “e-bike” rose faster than all stories **every year from 2021**, three years before CPSC created a separate e-bike product code in 2024.

Checked on its own, the old moped and power-assisted cycle code (where e-bikes were counted) also rose in 2021–2023. In the all-product scan those yearly numbers were too uncertain to pass the strict test, which is why the word check matters.

The scan also flagged e-scooter injuries in 2021 (+74%), the first year after their new code, and “water beads” in 2023.

Stories with e-bike wording, per year



Mentions of “e-bike”, “ebike”, “electric bike” and “electric bicycle” in sampled stories (a story can use more than one).

How the scan decides

- **Every product, every year.** Each year is compared with the year before (and the same months, for part-years).
- **Fair comparison.** A product counts only if it rose faster than all emergency visits, so a general rise (for example, a change in NEISS hospital weights) does not flag everything.
- **Real uncertainty.** Error ranges follow the NEISS hospital sample design (hospitals within size groups), including hospitals with zero cases.
- **Few false alarms.** About 800 products are tested at once, so a false-discovery control (Benjamini–Hochberg, 5%) is applied across all of them.
- **CPSC reporting rule.** Estimates with fewer than 20 cases, under 1,200 weighted visits or more than 33% relative error are not flagged.
- **Coding changes.** When CPSC splits or replaces a product code, the scan detects it and compares the old and new codes together.

What it does not do (yet)

- **National only, yearly by default.** A single month is much less sensitive: for e-bikes in 2025, one month needs about a 50% rise to be detected, a full year about 28%.
- **2024 shows no flags.** NEISS hospital weights rose sharply that year (some hospital-size groups by 27% and 46%), which hides product-specific rises.
- **2020–2021 are pandemic years.** Many sports and activity products swing up and down; read those flags with caution.
- **Coding practice can look like a hazard** (floors in 2025). Every signal needs a person to read the stories before acting.
- **Rates are per resident, not per rider.** Visits per 100,000 people show the burden on the population; risk per rider, trip or product sold needs exposure data and is the next step where public data exists.
- **Emergency visits only.** Urgent care, doctor visits and injuries never treated are not in NEISS, so all counts are a floor.

How to use it

Website: the Hazard Watch tab shows this year's signals and lets you download every table (CSV, with 95% ranges).

Python: `pip install -U neissai`, then three commands:
`neissai download` (public CPSC files)
`neissai hazard-export` (builds the input tables)
`neissai hazards` (writes the results, flag list, new-word lists and code-change table)

Ideas for safety and risk teams:

- Run it each year when CPSC releases new NEISS data.
- Use signals to decide which products to look at closely, not as final answers.
- Pair a signal with the matching injury stories and your own claims or incident data.

Validation so far

National totals match NEISS-AI's published estimates exactly (for example, bicycles in 2025: 420,962 visits). The code-change check finds exactly the three CPSC product code changes in 2019–2025 (scooters 2020, cribs 2021, e-bikes 2024). In a look back over 2020–2025, the scan flagged the e-scooter rise in 2021, the first year after e-scooters got their own code, and picked up e-bike wording from 2021, three years before e-bikes got theirs.

Data: U.S. Consumer Product Safety Commission, National Electronic Injury Surveillance System (NEISS), 2019–2025, public files; U.S. Census Bureau population estimates. Analysis: NEISS-AI, free and open source.
Source: NEISS-AI (Kumar, 2026), doi:10.5281/zenodo.23306744 · github.com/subash28kumar/neiss-ai